

UNIVERSITY OF WAH(UOW)
DEPARTMENT OF COMPUTER
SCIENCE



Lab Task # 12

COURSE: Computer Networks Lab_241L

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Registration NO: UW-24-CS-BS-101

Task

Adding Router to Screen



1841
Router0



1841
Router4



1841
Router1



1841
Router2



1841
Router3

Adding Computer to Screen



PC-PT
PC0



PC-PT
PC1



PC-PT
PC2



PC-PT
PC3



PC-PT
PC4



PC-PT
PC5



PC-PT
PC6



PC-PT
PC7



PC-PT
PC8



PC-PT
PC9



PC-PT
PC10



PC-PT
PC11



PC-PT
PC12



PC-PT
PC13



PC-PT
PC14

Adding Switches to Screen



2960-24TT
Switch5



2960-24TT
Switch6



2960-24TT
Switch7

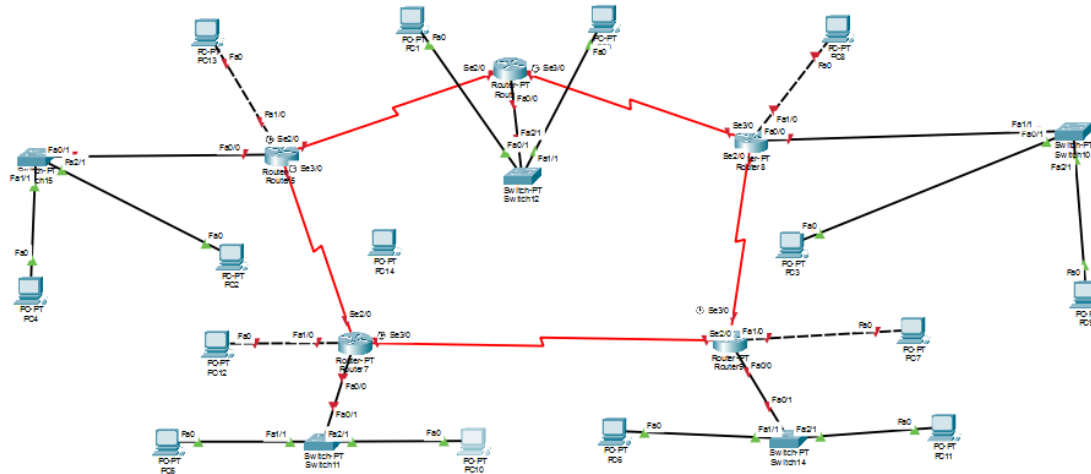


2960-24TT
Switch8

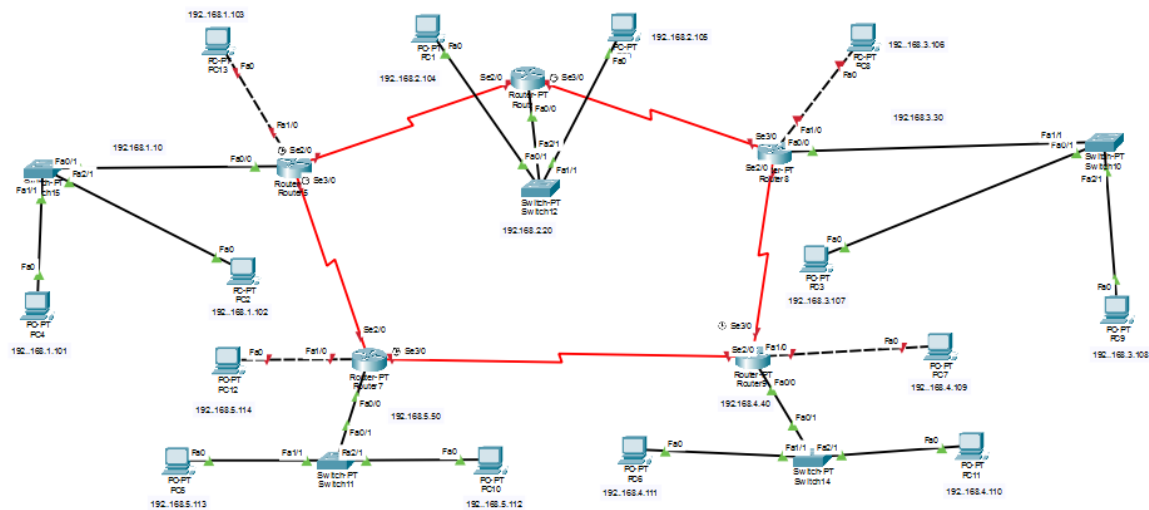


2960-24TT
Switch9

Arranging All as required and connecting according to give scienrio:



Giving IP address to all and Gateway PCs



IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.110.103

Subnet Mask

255.255.255.0

Default Gateway

192.168.110.111

DNS Server

0.0.0.0

Giving Router Address

Router 1

FastEthernet0/0	
Port Status	☒ On
Link Speed	☒ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
MAC Address	0001.9750.D0D1
IP Configuration	
IPv4 Address	192.168.1.10
Subnet Mask	255.255.255.0
Tx Ring Limit	10

I Give to 2 serial and 2 Fast Ethnet

FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0

RIP

RIP Routing

Network	
	Add

Network Address	
192.168.1.0	
192.168.10.0	
192.168.20.0	
192.168.110.0	

Router 2:

Serial2/0	
Port Status	☒ On
Duplex	☒ Full Duplex
Clock Rate	2000000
IP Configuration	
IPv4 Address	192.168.20.103
Subnet Mask	255.255.255.0
Tx Ring Limit	
	10

I Give to 2 serial and 2 Fast Ethnet

FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0

RIP

Network Address	
192.168.2.0	
192.168.10.0	
192.168.50.0	

Rouer 3:

FastEthernet1/0	
Port Status	☒ On
Link Speed	☒ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
MAC Address	0002.1735.B27B
IP Configuration	
IPv4 Address	192.168.130.113
Subnet Mask	255.255.255.0
Tx Ring Limit	10

I Give to 2 serial and 2 Fast Ethnet

FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0

RIP

RIP Routing	
Network	
Add	
Network Address	
192.168.3.0	
192.168.40.0	
192.168.50.0	
192.168.140.0	

Rouer 4:

FastEthernet0/0	
Port Status	☒ On
Link Speed	☒ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
MAC Address	0002.164A.3008
IP Configuration	
IPv4 Address	192.168.3.30
Subnet Mask	255.255.255.0
Tx Ring Limit	10

I Give to 2 serial and 2 Fast Ethnet

FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0

RIP

Network Address	Add
192.168.4.0	
192.168.30.0	
192.168.40.0	
192.168.130.0	

Rouer 5:

Serial2/0	
Port Status	☒ On
Duplex	☒ Full Duplex
Clock Rate	2000000
IP Configuration	
IPv4 Address	192.168.10.110
Subnet Mask	255.255.255.0
Tx Ring Limit	10

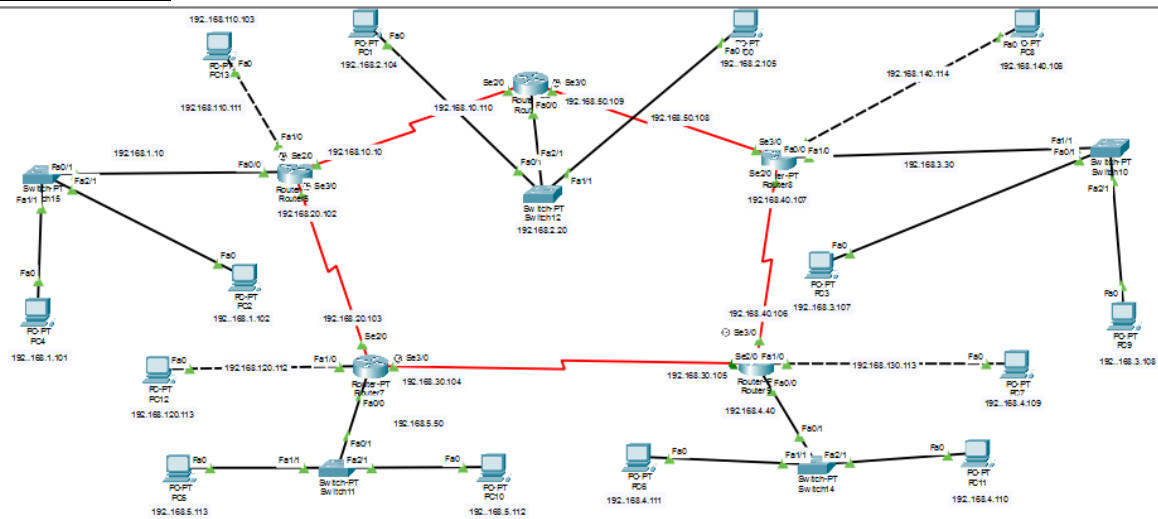
I Give to 2 serial and 2 Fast Ethnet

FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0

RIP

RIP Routing	
Network	Add
Network Address	
192.168.5.0	
192.168.20.0	
192.168.30.0	
192.168.120.0	
Remove	

Final Structure



Sending Messages

From Router 1 PC1 to All:

To router 2 PC 3

```
C:\>ping 192.168.110.103

Pinging 192.168.110.103 with 32 bytes of data:

Reply from 192.168.110.103: bytes=32 time<1ms TTL=127
Reply from 192.168.110.103: bytes=32 time<1ms TTL=127
Reply from 192.168.110.103: bytes=32 time<1ms TTL=127
Reply from 192.168.110.103: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.110.103:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

To router 3 PC 3

```
C:\>ping 192.168.120.113

Pinging 192.168.120.113 with 32 bytes of data:

Request timed out.
Reply from 192.168.120.113: bytes=32 time=2ms TTL=126
Reply from 192.168.120.113: bytes=32 time=11ms TTL=126
Reply from 192.168.120.113: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.120.113:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 11ms, Average = 4ms
```

To router 4 PC 1

```
C:\>ping 192.168.4.110

Pinging 192.168.4.110 with 32 bytes of data:

Request timed out.
Reply from 192.168.4.110: bytes=32 time=2ms TTL=125
Reply from 192.168.4.110: bytes=32 time=2ms TTL=125
Reply from 192.168.4.110: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.4.110:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms
```

From Router 2 PC3 to All:

To router 1 PC1

```
C:\>ping 192.168.1.101

Pinging 192.168.1.101 with 32 bytes of data:

Reply from 192.168.1.101: bytes=32 time=1ms TTL=126
Reply from 192.168.1.101: bytes=32 time=12ms TTL=126
Reply from 192.168.1.101: bytes=32 time=11ms TTL=126
Reply from 192.168.1.101: bytes=32 time=12ms TTL=126

Ping statistics for 192.168.1.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 12ms, Average = 9ms
```

To router 2 PC2

```
C:\>ping 192.168.3.107

Pinging 192.168.3.107 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.107: bytes=32 time=10ms TTL=126
Reply from 192.168.3.107: bytes=32 time=12ms TTL=126
Reply from 192.168.3.107: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.3.107:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 12ms, Average = 7ms
```

To router 3 PC1

```
C:\>ping 192.168.4.110

Pinging 192.168.4.110 with 32 bytes of data:

Reply from 192.168.4.110: bytes=32 time=2ms TTL=125
Reply from 192.168.4.110: bytes=32 time=12ms TTL=125
Reply from 192.168.4.110: bytes=32 time=2ms TTL=125
Reply from 192.168.4.110: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.4.110:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 12ms, Average = 4ms
```

To router 4 PC1

```
C:\>ping 192.168.1.101

Pinging 192.168.1.101 with 32 bytes of data:

Reply from 192.168.1.101: bytes=32 time=1ms TTL=126
Reply from 192.168.1.101: bytes=32 time=12ms TTL=126
Reply from 192.168.1.101: bytes=32 time=11ms TTL=126
Reply from 192.168.1.101: bytes=32 time=12ms TTL=126

Ping statistics for 192.168.1.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 12ms, Average = 9ms
```


From Router 3 PC1 to All:

To router 1 PC1

```
C:\>ping 192.168.2.104

Pinging 192.168.2.104 with 32 bytes of data:

Reply from 192.168.2.104: bytes=32 time=2ms TTL=126
Reply from 192.168.2.104: bytes=32 time=1ms TTL=126
Reply from 192.168.2.104: bytes=32 time=11ms TTL=126
Reply from 192.168.2.104: bytes=32 time=3ms TTL=126

Ping statistics for 192.168.2.104:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 11ms, Average = 4ms
```

To router 2 PC3

```
C:\>ping 192.168.5.112

Pinging 192.168.5.112 with 32 bytes of data:

Request timed out.
Reply from 192.168.5.112: bytes=32 time=4ms TTL=125
Reply from 192.168.5.112: bytes=32 time=13ms TTL=125
Reply from 192.168.5.112: bytes=32 time=3ms TTL=125

Ping statistics for 192.168.5.112:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 13ms, Average = 6ms
```

To router 1 PC1

```
C:\>ping 192.168.4.110

Pinging 192.168.4.110 with 32 bytes of data:

Reply from 192.168.4.110: bytes=32 time=2ms TTL=125
Reply from 192.168.4.110: bytes=32 time=12ms TTL=125
Reply from 192.168.4.110: bytes=32 time=2ms TTL=125
Reply from 192.168.4.110: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.4.110:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 12ms, Average = 4ms
```

To router 1 PC3

```
C:\>ping 192.168.1.101

Pinging 192.168.1.101 with 32 bytes of data:

Reply from 192.168.1.101: bytes=32 time=1ms TTL=126
Reply from 192.168.1.101: bytes=32 time=12ms TTL=126
Reply from 192.168.1.101: bytes=32 time=11ms TTL=126
Reply from 192.168.1.101: bytes=32 time=12ms TTL=126

Ping statistics for 192.168.1.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 12ms, Average = 9ms
```

To router 3 PC 3

```
C:\>ping 192.168.110.103

Pinging 192.168.110.103 with 32 bytes of data:

Reply from 192.168.110.103: bytes=32 time<1ms TTL=127
Reply from 192.168.110.103: bytes=32 time<1ms TTL=127
Reply from 192.168.110.103: bytes=32 time<1ms TTL=127
Reply from 192.168.110.103: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.110.103:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

To router 3 PC 3

```
C:\>ping 192.168.120.113

Pinging 192.168.120.113 with 32 bytes of data:

Request timed out.
Reply from 192.168.120.113: bytes=32 time=2ms TTL=126
Reply from 192.168.120.113: bytes=32 time=11ms TTL=126
Reply from 192.168.120.113: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.120.113:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 11ms, Average = 4ms
```

To router 4 PC 1

```
C:\>ping 192.168.4.110

Pinging 192.168.4.110 with 32 bytes of data:

Request timed out.
Reply from 192.168.4.110: bytes=32 time=2ms TTL=125
Reply from 192.168.4.110: bytes=32 time=2ms TTL=125
Reply from 192.168.4.110: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.4.110:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms
```